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REVIEW

Stromal vascular fraction: A regenerative reality? Part 2: Mechanisms of regenerative action

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Summary Adipose tissue is a rich source of cells with emerging promise for tissue engineering and regenerative medicine. The stromal vascular fraction (SVF), in particular, is an eclectic composite of cells with progenitor activity that includes preadipocytes, mesenchymal stem cells, pericytes, endothelial cells, and macrophages. SVF has enormous potential for therapeutic application and is being investigated for multiple clinical indications including lipotransfer, diabetes-related complications, nerve regeneration, burn wounds and numerous others. In Part 2 of our review, we explore the basic science behind the regenerative success of the SVF and discuss significant mechanisms that are at play. The existing literature suggests that angiogenesis, immunomodulation, differentiation, and extracellular matrix secretion are the main avenues through which regeneration and healing is achieved by the stromal vascular fraction.

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Introduction

The concept of adipose tissue as a reservoir of regenerative cells has gained widespread interest after adipose derived stem cells (ADSCs), a form of mesenchymal stem cells (MSCs), were characterized by Zuk et al. in 2001.[1] ADSCs, which are easily extracted from adipose tissue, have been intensely studied due their multipotent differentiation capacity, surface markers and adherence to plastic.[2] A recent shift of focus has directed the attention from the study of ADSCs to that of a heterogeneous mixture of cells from which they are derived, the stromal vascular fraction (SVF).[2,3]

Despite the burgeoning research on these two populations of cells, few studies compare the therapeutic effects of ADSCs with SVF cells. In those that do, SVF treatment provided similar therapeutic effects to ADSC treatment, as seen in osteochondral defects and myocardial infarction therapies.[4,5] In experimental autoimmune encephalitis studies, SVF demonstrated similar neuroprotective effects and greater immunomodulatory properties than ADSCs.[6] These studies indicate that SVF treatment is not only comparable, but in some cases better than ADSC treatment.

The growing research on SVF has been validated by its use in several therapeutic models, including radiation therapy, retinopathy and nerve regeneration.[7–9] When applied to such models, SVF demonstrates angiogenic, immunomodulatory, differentiation, and extracellular production qualities that are important in regeneration and repair.

The regenerative capacity of SVF is likely derived from the heterogeneity of its constituents that provide numerous mechanisms for regeneration to occur. SVF is a source of progenitors and stem cells, which have the potential to differentiate along different lineages. Numerous studies have previously demonstrated the ability of cells found in the SVF,[1,10,11] including M2 macrophages,[12] to differentiate into osteogenic, adipogenic, chondrogenic cell types.

One of the most abundant cell types is the preadipocyte, the precursor to the mature adipocyte. Recent evidence suggest that this cell, also described as a supra-adventitial adipose stromal cell or dedifferentiated adipose cell, shares many of the same phenotypic markers and characteristics of MSCs, implicating its involvement in regeneration.[13,14] SVF also contains endothelial progenitor cells (EPCs), which have the capacity to induce angiogenesis through the release of growth factors such as vascular endothelial growth factor (VEGF) and insulin-like growth factor-1 (IGF-1).[15] The macrophages and monocytes found in SVF have been shown to mediate the immune response through the expression of various cytokines,[16] and some exhibit plastic adherence and the ability for multilineage differentiation seen in ADSCs.[12] These macrophages are modulated by T regulatory cells, which can have immunosuppressive properties.[17] Additionally, pericytes found in adipose-derived SVF have been demonstrated to regenerate muscle tissue when injected into damaged mouse muscle,[18] and stromal cells can secrete extracellular matrix components that may improve the general capacity for cellular adhesion, migration, cell–matrix interaction and regeneration.[19,20]

The aim of this review is to present and discuss the cellular and molecular mechanisms behind the regenerative properties observed with stromal vascular fraction. We have included an overview of the surface markers important to SVF cellular interaction, as well as sections addressing the regenerative mechanisms and theories surrounding the tissue survival sequence following SVF implantation.

Cell surface markers (Table 1)

Cell surface molecules, including the clusters of differentiation (CD), are crucial for cell–cell and cell–environment interactions, but are also used to define this mixed population. Despite position papers from the International Federation for Adipose Therapeutics and Science (IFATS) and the International Society of Cellular Therapy (ISCT), there is little consensus with regards to CD characterization of the cells that comprise the SVF. We have identified the markers that are most commonly cited in recent literature specifically related to the regenerative component of uncultured human SVF cells (Table 1).

Table 1 Stromal vascular fraction regenerative cell surface markers recently defined in the plastic surgery literature.

	Mesenchymal stem cells (ADSCs)	Pericytes	Endothelial progenitor cells	Supra-adventitial adipose stromal cells (Preadipocytes)	Transitional cell	Macrophages/ monocytes
Bourin et al., 2013[2]	$CD45^{-}CD31^{-}CD13^{+}CD73^{+}CD90^{+}CD105^{+}$ $CD34^{+}$ (unstable)					
Zimmerlin et al., 2013[14]	$CD45^{-}$ (<50%) $CD31^{-}$ (<20%) [$CD13^{+}CD29^{+}CD44^{+}CD73^{+}CD90^{+}$ (>40%)] $CD34^{+}$ (>40%)	$CD45^{-}CD31^{-}CD146^{+}$	$CD45^{-}CD31^{+}CD34^{+}$	$CD45^{-}CD31^{-}CD146^{-}CD34^{+}$	$CD146^{+}$ $CD31^{-}CD34^{+}$	
Hager et al., 2013[25]			$CD133^{+}CD146^{+}$ $CD34^{+}CD31^{+}$			
Bianchi et al., 2013[65]	$CD90^{+}CD29^{+}$ $CD34^{-}$	$CD146^{+}CD90^{+}$ $CD34^{-}$			$CD146^{+}$ $CD34^{+}$	
Corselli et al., 2013[24]		$CD146^{+}CD34^{-}CD45^{-}CD56^{-}$	$CD34^{+}CD146^{+}$ $CD31^{+}$	$CD34^{+}CD31^{-}CD146^{-}$		
Traktuev et al., 2008[66]	$CD34^{+}CD31^{-}CD45^{-}CD144^{-}$			$CD34^{+}CD45^{-}CD31^{-}$		
Eto et al., 2013[12]						$CD45^{+}CD14^{+}$ $CD34^{+}CD206^{+}$

Positivity (+) is defined as greater than 80% unless otherwise noted. Negativity (−) is defined as less than 5% unless otherwise noted.

According to Bourin et al., the stromal elements of the SVF (20%), which excludes endothelial (10–20%) and hematopoietic cells (25–45%), are commonly characterized by the combination $CD45^{-}CD235^{-}CD31^{-}CD34^{+}$.[2] This group went further by setting forth guidelines for the characterization of SVF using the markers CD13, CD29, CD44, CD73, and CD90 (>40% expression) as well as CD34 (>20% expression).[2] Unfortunately the sensitivity and specificity of these marker expression levels as they pertain to the regenerative potential of this cellular mix are not clear. Interestingly, while typical markers for mature ADSCs (e.g. CD13, CD73, CD90 and CD105) are also observed in this immature SVF population (to a lesser degree), a high expression of both CD13 and CD73 on the $CD45^{-}CD31^{-}CD34^{+}$ population of cells is specific for a mesenchymal origin and implicates these primitive cells in regenerative processes. This finding is supported by additional studies.[14,21]

Ironically, CD34 was originally described as a primary negative marker for MSC identification by the ISCT in 2006.[22] Since that time, numerous studies have demonstrated the presence of CD34 on freshly isolated MSCs that is lost with successive culture.[2,23] This observation was reflected in the revised joint statement by the ISCT and IFATS in 2013.[2] Thus, considerable attention has been turned to its expression, a marker that is closely linked with progenitor activity and perivascular tissue.[21,23]

Recently, two groups have described a population of perivascular cells in the SVF that are defined by $CD31^{-}CD45^{-}CD146^{-}CD34^{+}$[14,24] (Table 1). Again, this population was found to be strongly positive for the typical MSC markers, and Zimmerlin et al. demonstrated that this population represented 59% of the total SVF isolation.[14] They suggest that these cells, usually described as preadipocytes, be termed supra-adventitial adipose stromal cells (SA-ASCs) due to the high likelihood that they also possess significant stem cell-like properties.[14]

Other important perivascular subtypes with progenitor activity include EPCs and pericytes (Table 1). Endothelial progenitor cells can be defined by a combination that includes $CD45^{-}CD31^{+}CD34^{+}CD146^{+}$.[14,24,25] Pericytes, which are also closely associated with walls of the vessels that intimately invest the adipose tissue, express $CD45^{-}CD31^{-}CD146^{+}$.[14,24,25] Zimmerlin et al. demonstrated through immunohistochemistry and flow cytometry that each of these population of cells possess high proliferative potential (EPCs > pericytes > SA-ASCs) and expression of typical MSC markers (AS-ASCs > EPCs > pericytes).[14] Interestingly, they also identified a transitional population of cells ($CD31^{-}CD34^{+}CD146^{+}$) that likely serves as an intermediate between these three populations and exhibits superior proliferative and mesenchymal-like phenotypic potential.[14]

Finally, the macrophage is emerging as an important immunological cell found in the SVF. In their early study, Astori et al. recognized that 10.9 ± 9.6% of the SVF cells are positive for the monocytic marker CD14.[10] It was recently demonstrated that $CD14^{+}$ monocytes found in the SVF likely contribute to tissue vascularization via the secretion of cytokines such as VEGF and basic fibroblast growth factor (bFGF).[26] Most notably, Eto et al. recently discovered a subpopulation of the anti-inflammatory M2 phenotype adipose tissue resident macrophages ($CD45^{+}CD14^{+}CD206^{+}$)

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that are CD34^{+} and possess MSC-like characteristics including plastic adherence, morphology and pluripotency,[12] thus implicating these cells in the regenerative potential of the SVF (Table 1).

Components of regeneration

While SVF cells may contribute to tissue formation by differentiation of their individual components, there is also evidence of regeneration occurring through paracrine signaling involving cross-talk between the different cell populations in SVF and their host environment.[27] Blaber et al. demonstrated that SVF cells are involved in extensive paracrine signaling when co-cultured with adipocytes, with increased cytokine production noted *in vitro*.[27] Chazenbalk et al. provided further evidence of paracrine activity by co-culturing adipocytes with adipose-derived macrophages and stem cells to induce preadipocyte formation.[28] Hence, evidence supporting the potential benefits of cellular crosstalk amongst SVF components is well supported by the generation of new cells and the secretion of chemical mediators. Conversely, the behavior and differentiation of SVF cells can also be influenced their local milieu. These interactions between different SVF components and their host tissue bring about regenerative changes secondary to angiogenic, immunomodulatory, differentiation and stromal inductive mechanisms.

Angiogenesis

The ability of SVF to promote angiogenesis and neovascularization has major implications for diseases characterized by poor vascularization, ischemia, and necrosis. Its application has successfully resulted in neovascular formation when applied to acute myocardial infarction, cosmetic procedures, burn wounds, diabetic foot ulcers and ischemic muscle.[4,29–34] For instance, in a rat model of second-degree burns that were treated with SVF, researchers noted increased levels of VEGF and higher vascularization when compared to the control group.[32] The angiogenic properties of SVF appear to be a function of its heterogeneous cellular make up. This includes several cell populations that, when combined, facilitate the formation of new blood vessels, including adipose stem cells, EPCs, macrophages, and fibroblasts.

In that regard, co-implantation of EPCs and ADSCs resulted in greater neovascularization than EPCs or ADSCs alone *in vitro*.[21] Others went further to demonstrate that the cellular components of SVF are rapidly reassembled to form new vessels in the recipient tissue.[35] The process of neovascularization is further stimulated by stromal cells through the release of growth factors such as vascular endothelial growth factor (VEGF), hepatocyte growth factor (HGF) and transforming growth factor-β (TGF-β).[36] Koh et al. showed that macrophages found in SVF are also important for the proper structural organization of new vessels.[35]

In the setting of lipotransfer, fat graft survival is unpredictable with resorption rates that can exceed 75% of the originally grafted material.[37] As a form of cell-based therapy to counteract this resorption, SVF enrichment promotes secretion of a diverse array of cytokines and growth factors that encourage angiogenesis and fat graft revascularization.[27] In particular, pericytes and endothelial cells directly contribute to vessel formation and angiogenesis.[38] Additionally, MSCs have been demonstrated to secrete VEGF, HGF, and transforming growth factor beta 1 (TGF-β1), thus contributing to angiogenesis through a paracrine effect.[39]

Recent animal studies are aiding in the characterization of the biological sequences that SVF cell enrichment contributes to fat graft survival.[40,41] Zhu et al. established that SVF cell-assisted lipotransfer (CAL) in nude mice resulted in larger graft volume with enhanced angiogenesis and morphology[40]; in contrast, the grafts in the control group appeared less viable with fibrous encapsulation. They noted rapid growth of a vascular network in the SVF enriched group that gradually extended from the peripheral to central regions of the adipose tissue; concurrently, there was a significant increase in VEGF expression observed in the SVF group,[40] which can influence the migration of endothelial and stromal cells to the neovascularized region. New adipocytes demonstrated a similar tendency to gather around the blood vessels in the interstitial tissue, indicating that the adipocyte likely depends on this vascular network for its viability.

Recently, a study by Paik et al. revealed that it is also important to consider the concentration of SVF cells implanted with the graft. In this mouse model, SVF was mixed with lipoaspirate at concentrations ranging from 10,000 to 10 million cells per 200 μL. Their results indicated an inverse dose-dependent relationship, in which higher cell concentrations interfered with graft retention via inflammation and low vascularity. These findings suggest that spatial arrangements and availability of nutrients for the implanted cells must be factored into our understanding of the SVF regenerative process.[42]

SVF can also be used to create highly vascularized tissue engineered human dermo-epidermal skin substitutes (DESS), a treatment for burn wounds.[43] DESS are usually not prevascularized, and therefore depend on diffusion during the first few weeks after transplant to survive. Immune deficient rats treated with DESS supplemented with SVF isolated from human adipose tissue displayed vascularized tissue anastomosed to the rat vasculature within four days, efficiently establishing tissue homeostasis. The composition of the capillaries in the engineered DESS was similar in nature to normal human capillaries. The mechanisms through which SVF form capillary plexuses are still unclear but may include secreting growth factors such as VEGF and IGF. Due to the vascularization present in the tissue, the dermis underwent almost no contraction. Furthermore, freshly isolated SVF had greater vasculogenic properties when compared to cultured SVF.[43]

Angiogenesis has also been shown to improve the healing process in cardiac disease. Premaratne et al. transplanted SVF and bone marrow mono-nuclear cells (BM-MNC) in rats with moderate myocardial infarction (MI) via intramyocardial injection into the chronically ischemic myocardium.[34] Vascular density in the groups treated with SVF and BM-MNC were significantly higher than in the control group, and the SVF group produced the greatest vascular density of vessels: larger than 30 μm in diameter in the peri-MI area

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demonstrating the superior angiogenic capabilities of SVF. Both SVF and BM-MNC groups showed a lower percentage of fibrotic tissue inside the infarct area when compared to control groups, as well as reduced left ventricular dilation and an improvement in cardiac function.[34]

Rigotti et al. postulated a common sequence of events occurring when SVF is transplanted to radio-damaged tissue.[8] This involves the targeting of damaged tissue, vascular growth factor release, neovascularization, and tissue oxygenation which ultimately results in reperfusion and recovery of the damaged tissue.[8] Similar sequences of angiogenic events were demonstrated in other disease models, suggesting that SVF has the potential to be used for therapeutic neovascularization and tissue repair across a spectrum of conditions. For instance, in bone fractures and defects, SVF can secrete VEGF to encourage the formation of new networks of capillaries and recruit hematopoietic stem cells to form new bone.[44] Similarly, Koh et al. implanted SVF directly onto ischemic murine muscle and observed blood perfusion and the formation of new vascular networks in the muscle.[35] These results are illustrative of SVF's ability to act as a versatile angiogenic agent in a variety of disease models.

Immunomodulatory effects

Many of the therapeutic studies have described an initial decrease in inflammation and immune response at the site of SVF injection.[41] The immunomodulatory properties of SVF again relate to its diverse cellular components. Stem cells and progenitors are known to have anti-inflammatory and anti-apoptotic properties that contribute to regeneration of host tissue. Macrophages in healthy adipose tissue are commonly of the anti-inflammatory M2 phenotype, some subpopulations of which have demonstrated differentiation capacity.[12,16] SVF also contains T-regulatory (Treg) cells that express high levels of immune suppressive cytokines. These Treg cells can maintain the M2 phenotype of the macrophages in SVF when appropriate.[17]

When SVF is applied to certain disease models, it tends to decrease inflammatory cytokines and growth factors such as tumor necrosis factor alpha (TNFα) and interleukin 6 (IL-6).[34] In an experimental autoimmune encephalitis model for multiple sclerosis, SVF was even shown to be more effective than ADSCs at reducing inflammatory cytokines and growth factors such as interferon-γ and IL-12. Both are important cytokines that proliferate as part of the pathogenesis of multiple sclerosis.[6] Thus, SVF mediates inflammation and the immune response through the expression and suppression of various cytokines. This further supports the notion that SVF facilitates regeneration both directly through cytokine release and indirectly through paracrine effects on the host tissue.

Regenerative capacity: differentiation and proliferation

Stem cells are an appealing regenerative choice due to their natural ability to differentiate into specific lineages when placed in the appropriate environment. Aside from this multipotency, SVF can also induce host cell proliferation when introduced to fat tissue, diabetic foot ulcers, nerves and burn wounds.[4,29–34,45,46]

In addition to SVF acting in a paracrine role to improve fat graft retention during CAL, cells contained therein can also replace adipocytes and reduce the load of volume loss. Zhu et al. proposed that primitive MSCs from the SVF could differentiate into neonatal adipocytes after observing CD34 expression in the graft.[40] Using the premise that CD34 is a MSC and immature cell marker, the authors initially detected CD34 on early stage spindle cells. Fourteen days later, small CD34+ neonatal adipocytes were found, an indication they had differentiated from the original spindle cells. At 30 days, even more CD34+ neonatal adipocytes were found in various differentiation stages. The authors concluded that MSCs from the SVF can regenerate adipose tissue.[40] In another study, Fu et al. used green fluorescence protein-positive SVF to illustrate the dynamic changes that occur in mice treated with CAL.[41] The SVF cells not only survived in the fat graft but also differentiated into adipocytes.

SVF increases the proliferation of fibroblasts when injected into diabetic foot ulcers, thus aiding the healing process by 'awakening' these senescent cells.[46,47] A similar result has been observed in regard to burn wounds in which SVF administered through intradermal injections helped increase fibroblastic activity and cellular proliferation.[32]

The specific mechanisms through which SVF promotes regeneration of nerve cells are not conclusively defined, but studies using SVF for nerve repair have shown that animals treated with SVF exhibit faster recovery of regenerated axons and a greater number and diameter of myelinated fibers.[45,48] This nerve regeneration must either be a result of the differentiation of SVF stromal cells or their paracrine effect on host tissue, which could induce the surrounding Schwann cells to regenerate and migrate. Both mechanisms are likely to contribute to this process.

While certain SVF components have the capacity to proliferate and differentiate into mature cell types, it is important to acknowledge that ADSCs have similar capabilities. Given that ADSCs are derived from plastic-adherent populations within the SVF, the similarity of their regenerative properties is not unexpected.

Extracellular matrix (ECM) components

The extracellular matrix acts as a potent scaffold in many tissue types, accelerating the regenerative functions of nearby cells.[49] It is comprised of structural proteins such as collagen, laminin, fibronectin, and elastin, which are commonly secreted by the fibroblast. One of the hallmarks of ECM function is its ability to interact dynamically with integrin proteins on adhesive cells. The attachment of integrins to ECM receptors triggers signaling pathways and changes in cell activity.[50,51] The ECM also facilitates cellular movement and migration, as the cell uses integrin-cytoskeleton attachments to pull itself forward.[52,53] Furthermore, the ECM contributes to the growth of vascular networks by mediating morphogenesis and migration speeds during angiogenesis.[54] Since the SVF contains matrix-secreting fibroblasts and other stromal cells, the application of SVF is potentially advantageous to tissue types that benefit from a three dimensional (3D) scaffold.

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In lipotransfer, a number of synthetic and human-derived scaffolds (often in the form of gels or powders) have been injected in various settings to improve volumetric retention.[55] Recently, Debels et al. showed that adipose-derived acellular matrix (hDAM) alone was just as effective as fat in promoting fat regeneration, however fat supplemented with hDAM was superior to both.[56] Hence, it appears that the presence of a 3D structure facilitates the proliferation of nearby adipocytes.[56,57] It is likely that collagen-secreting fibroblasts from the SVF improve fat graft viability during cell-enriched lipotransfer via similar mechanisms.

Nerve defects treated with SVF may also benefit from the combination of ECM components. Mohammadi et al. showed that a conduit filled with SVF accelerated axon recovery and increased the number and diameter of myelinated fibers.[9] One plausible explanation is that ECM proteins (e.g. laminin and collagen) secreted by the SVF facilitates the attachment, adhesion and migration of Schwann cells along the basal lamina. Laminin can even mediate Schwann cell behavior, such as proliferation, radial sorting and myelination.[58] The summation of these ECM components may significantly enhance neurite outgrowth during transections.

Skin grafts are commonly used as a form of treatment for burn wounds. When SVF cells are combined with the skin graft and injected into the wound bed, fibroblasts can secrete ECM proteins to form a scaffold that accelerates healing. The presence of anti-inflammatory M2 macrophages can then play a crucial role in the dynamic remodeling of the matrix.[59] In contrast to many synthetic matrices, SVF injection represents a natural vehicle for creating a regenerative microenvironment.

Survival sequences of SVF implantation

There are a few conventional theories concerning the survival sequences that occur during lipotransfer. In the host replacement theory, macrophages phagocytize lipid droplets changing their form from a spherical cell into a more robust, flat, spindle shaped, fibroblast-like phenotype that is more capable of surviving an ischemic environment. This process of dedifferentiation is a survival mechanism for these adipocytes which can then redifferentiate when the environment improves.[40] Similarly the process of autophagy (fragmentation of the cell) facilitates survival of the cells by recycling usable fragmented components.[40] Either way, the process allows the cells to survive the initial period of hypoxia and increased inflammation and prepares the ECM milieu for one of regeneration by inducing vascularization and immune modulation.[40,41] Once inflammation is controlled and vascularity is established, the regenerative process can take place from exogenous cells transplanted to the area via the graft, and/or from surrounding endogenous host tissue that has been influenced by SVF components.

The cell survival theory claims some transplanted cells can simply survive for extended periods after injection. However, based on the results of their study, Zhu et al. proposed that the survival of adipocytes in the graft depends on the synergistic mechanisms of angiogenesis, differentiation and immunomodulation.[40] Adipocytes, in large part, depend on the presence of a nearby vascular network to survive. SVF growth factors like VEGF and HGF contribute to angiogenesis. Tracking CD34 markers showed that stromal stem cells from the SVF differentiate into mature adipocytes. Furthermore, migration from host tissue and dedifferentiation can also supply progenitors. The subsequent crosstalk between progenitors and immune cells help mediate the microenvironment for regeneration.[40,41] Thus regeneration appears to be a staged process of wound milieu adjustment followed by interactions between components of the grafted tissue with the wound bed and with cellular components of the host tissue.

Discussion

While ADSCs have been heavily investigated as a regenerative modality over the past decade, recent evidence suggest that the SVF may be more advantageous from a clinical standpoint, a byproduct of its quicker isolation and heterogeneity. With its stromal, multipotent and hematopoietic cell populations, SVF has the capacity to achieve regenerative success through multiple cell components. Just as pericytes and endothelial cells can synergistically develop new blood vessels, anti-inflammatory immune cells can mediate the local milieu and foster an optimal regenerative microenvironment. Hence, while few studies compare the mechanisms through which SVF and ADSCs achieve regeneration, we believe that the heterogeneity of the former is more conducive to healing and repair and offers significant translational advantages.

Endothelial cells and pericytes are thought to be closely intertwined in the formation, stabilization and remodeling of vascular networks.[38] Much of their coordination arises from the release of factors such as TGF-β and PDGF-β. Through these molecular signaling pathways, endothelial cells and pericytes can stimulate reciprocal recruitment and proliferation during angiogenic processes. In fact, *in vivo* models have showed that the absence of pericyte–endothelial interactions leads to a number of pathological conditions.[60] In addition to their role in angiogenesis, pericytes have been postulated to have stem-cell like properties, including the capacity to differentiate into adipocytes, chondrocytes, osteoblasts and granulocytes. This has led to their proposed characterization as MSCs, with which they share similar surface markers.[60,61] Thus the initial obsession with the homogenous generation of ADSCs has found an alternative in a heterogenous collection of progenitor cells that have similar capabilities of differentiation, growth factor elaboration and paracrine influences.

Aside from their capacity to differentiate into mature cell types, the stromal population within the SVF, including fibroblasts and stem cells, has been shown to stimulate angiogenesis and secrete important ECM proteins like collagen.[62] Newman et al. suggested that fibroblast-derived extracellular matrix components are crucial for lumen formation during the development of blood vessels and that angiogenesis necessitates synergy between stromal and endothelial populations.[63] Using RNA technology and purified protein, they demonstrated that stromal cell proteins such as hepatocyte growth factor and fibronectin were conducive to the sprouting of endothelial cells.[64] Thus regenerative mechanisms such as angiogenesis and ECM-protein synthesis are interdependent and synergistic in establishing a suitable milieu for healing.

Regarding inflammation, Bourin et al. reported that 5–15% of SVF derived from healthy adipose tissue is comprised of monocytes/macrophages, many of which are potentially of the anti-inflammatory M2 phenotype, an important component in controlling the environment for regeneration.[2] In addition, 10–15% of the SVF is comprised of lymphocytes, which includes regulatory T cells[2] that can monitor the immune response accordingly at the wound site. Thus the interplay of controlled inflammation and immune modulation of the wound milieu as a precursor to angiogenesis, ECM formation, and finally cellular regeneration provide the necessary constituents for wound bed preparation and subsequent healing.

In theory, the summation of these various cell types appears to indicate that there are advantages to using a heterogeneous cell mix. Although SVF is a more clinically implementable option than ADSCs, it is important to acknowledge that ADSCs may be more a more potent representative of a defined regenerative cell than any single component of SVF. However when considered as a combined regenerative mix, SVF offers a range of options that encourage crosstalk and synergy between its own components and that of the host tissue.

While there is still uncertainty regarding the phenotypic identification of SVF components using CD markers, recent studies suggest there is an intimate relationship of perivascular cells exhibiting the markers CD31, CD34 and CD146, all of which exhibit progenitor activity. Of the markers, CD34 appears to be the most important identifier with regards to this regenerative activity.

Conclusion

The numerous cell types in SVF, including ADSCs, macrophages, and endothelial progenitor cells, have distinct contributions to the regenerative capabilities of SVF. We propose that the heterogeneous nature of SVF provides properties of immunomodulation, controlled inflammation, angiogenesis, differentiation and extracellular matrix production to promote regeneration. While these properties are clearly present in SVF-induced regeneration, the exact mechanisms and contributions of individual cell populations are yet to be elucidated. Additionally, alternatives techniques for non-enzymatic harvesting of SVF (NESVF) are being sought. Overall, the concept of a regenerative mix derived from lipoaspirate with minimal manipulation is an extremely appealing possibility for future therapeutic options.

Conflict of interest statement

The authors declare no potential conflicts of interest with respect to research, authorship and publication of this article.

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